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The knowledge of sleep medicine among clinical dental students at Case Western Reserve University

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Objective. The aim of this study is to evaluate the knowledge of sleep medicine among dental students to assess if more focus needs to be put on this topic as part of the students' education process.

Study Design. A cross-sectional study was conducted to evaluate knowledge of sleep medicine in 70 third- (D3) and fourth-year (D4) dental students at Case Western Reserve University in Cleveland, Ohio. Participants completed an electronic Assessment of Sleep Knowledge in Education (ASKME) questionnaire.

Results. The mean score of the ASKME questionnaire was 14.8 ± 4.8. There was a highly statistically significant correlation between gender and mean number of correct answers, but no correlation between the number of correct answers and the current year of dental school. Overall, 67.1% of participants had low knowledge on the topic of sleep medicine.

Conclusions. This study demonstrates that dental students have a low level of knowledge about sleep medicine and sleep disorders. Dental faculties should incorporate courses related to sleep medicine into the dental school curriculum. (Oral Surg Oral Med Oral Pathol Oral Radiol 2026;000:1−5)

Sleep disorders are a prevalent health concern across the world. The International Classification of Sleep Disorders notes more than 80 different disorders, including obstructive sleep apnea (OSA), narcolepsy, and periodic limb movement disorder.[1] In America, it is estimated that between 50 and 70 million people are affected by sleep disorders.[2] OSA is the most common sleep disorder, with a 2006 study estimating that OSA affects 5%-10% of populations (approximately 18 million Americans).[3] OSA consists of repeated narrowing and collapsing of the throat during sleep leading to "intermittent hypoxia and hypercapnia, recurrent arousals and increase in respiratory efforts, leading to secondary sympathetic activation, oxidative stress and systemic inflammation" in addition to other comorbidities.[4]

Comorbidities associated with sleep disorders include: hypertension, coronary artery disease, stroke, and diabetes.[5] Despite the prevalence of sleep disorders and their adverse effects on health and quality of life, sleep medicine is still a largely unexplored medical field worldwide. Many studies on the topic note that sleep disorders are often under-diagnosed by primary care physicians and unrecognized by the general public.[6,7] A 2001 study found that the majority of primary care physicians (75.2%) were not routinely screening for sleep disorders during visits, and significantly less were referring patients to sleep or psychiatry clinics.[8] A 2013 study furthered this finding by discovering that many general practitioners did not understand the impact of sleep disorders on their patients' lives or see the value in consulting with sleep specialists.[9]

Dentists play a crucial role in identifying and diagnosing sleep disorders. Many studies have found that oral and general exams can be used to diagnose OSA.[10] Neck circumference, tongue size, Mallampati score, and palatal vault depth can all be assessed by dentists to determine if a patient should be referred to a sleep physician.[10] Additionally, diagnostic tools like lateral cephalometric radiographs which evaluate soft palate size, airway space, and craniofacial relationships, and Cone Beam CTs can be used to diagnose OSA.[11]

Studies in America and worldwide have found that insufficient time is being dedicated to sleep medicine in the medical school curriculum. A 1993 study found that an average time of less than 2 hours was being spent on sleep medicine, with over $^2/_3$ of study participants describing their education on this topic as "inadequate," while a 2002 study found that the average time spent on sleep medicine education increased to 2.1 hours.[12,13] A 2008 study found that the average time spent on sleep medicine had increased to 3.92 hours, with the main topics of education being sleep-related breathing problems and sleep bruxism.[14] Studies in Jordan and Saudi Arabia found that the average time spent on sleep medicine education in medical schools was 2 and 3.6 hours, respectively.[5,15]

Statement of Clinical Relevance

This study evaluates the knowledge of sleep medicine among dental students. The goal of this study is to assess if more focus needs to be put on this topic as part of the students' education process.

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While similar studies have been conducted to assess sleep medicine education and knowledge worldwide, it is important for these studies to be done regularly to assess the students' knowledge and to identify opportunities to improve the curriculum. Our study chose to focus on dental students because dentists play a unique role in identifying and diagnosing sleep disorders, and this topic is still largely unexplored.

The aim of this study is to evaluate the knowledge of sleep medicine among dental students to assess if more focus needs to be put on this topic as part of the students' education process. We hypothesize that dental students have insufficient knowledge of sleep medicine, likely due to minimal coverage of this topic throughout their curriculum.

The primary outcome variable was the total ASKME questionnaire score, calculated as the number and percentage of correctly answered questions. The secondary outcome variable was performance within the subcategories of high or low knowledge based on these scores. Predictor variables were gender and dental school year. While the ASKME questionnaire does not have a formally defined measure of what score indicates knowledge on this topic, we considered 18 correct answers (≤60% correct) to be the threshold for knowledge in this study, as this was the measure used in previous studies on the topic.

MATERIALS AND METHODS

This cross-sectional study was conducted during the 2024-2025 academic year. The study population was third- and fourth-year dental students at Case Western Reserve University. The total number of participants was 70 (33 male, 35 female, 2 preferred not to say). This study was approved by the Case Western Reserve University Institutional Review Board.

ASKME questionnaire

This study was conducted using the Assessment of Knowledge in Sleep Education (ASKME) questionnaire, a "standardized assessment measure for use in medical education in sleep," which was created in 2001 and has been utilized in many accredited studies since that time.[16] An email containing a link to the online ASKME questionnaire was sent to all clinical dental students at Case Western Reserve University from the department administrator. This survey was conducted via the Qualtrix survey platform. Participants consent was granted prior to accessing the survey. The survey included 34 questions: 4 questions regarding demographic information and 30 questions to assess their knowledge on sleep medicine. These 30 sleep medicine questions spanned 4 categories: sleep physiology and sleep stages (9 questions), sleep and age (5 questions), sleep disorders and abnormalities (14 questions), and sleep and medical conditions/pharmacology (2 questions). All questions were answered with "true," "false," or "I don't know." As participants had the option to skip any questions as well, we only included survey data from surveys that had over half of the questions answered with one of the above options. The ASKME questionnaire questions can be referenced in Table I.

Data analysis

Survey data was analyzed to find the mean number and percentage of each survey question answered correctly. Independent sample t tests were used for continuous data, expressed as mean ± standard deviation and Chi-square tests were used for categorical data, expressed as absolute numbers (n) and percentages. The P value was derived to determine if these results were statistically significant ($P \leq .05$).

RESULTS

Study participants ranged in age from <26 years old to ≥27 years old. A total of 40 D3 students (57%) and 30 D4 students (43%) participated in this study. Survey data were gathered from 35 female students (50%), 33 male students (47.1%), and two students who did not identify their gender (2.9%). The students who did not identify their gender were excluded from the independent sample t test comparing means by gender and from Chi-square test comparing knowledge groups based on gender.

In scoring the data, only correct answers were counted as the participant being knowledgeable on the topic. Incorrect answers, answers of "I don't know," and skipped questions were counted as the participant not being knowledgeable on the topic. The mean score of correctly answered questions was 14.8 ± 4.8 (49.3%). The mean number and percentage of correct answers for each survey question can be referenced in Table I. A total of 47 out of 70 participants (67.1%) answered less than 60% of questions correctly, placing them in the low knowledge category.

Female students answered 16.4 ± 3.9 questions correctly, while male students only answered 13.2 ± 5.0 questions correctly. There was a highly statistically significant difference between mean scores by gender ($P < .0001$). A total of17 female participants (48.6%) were in the high knowledge group, as opposed to 5 male participants (15.1%). Chi-square test revealed a statistically significant correlation between genders ($P = .0013$).

D3 students answered 14.4 ± 5.0 questions correctly, and D4 students answered 15.3 ± 4.6 questions correctly. There was no statistically significant difference between dental school year and the mean number of correctly answered questions on the ASKME

Table I. The mean number and percentage of each question answered correctly by study participants

No.	*ASKME questionnaire*	n *(%)*
1	The need for sleep decreases in persons above 50 years of age.	29 (41.4)
2	Melatonin is a natural body hormone that typically increases during nighttime hours.	63 (90.0)
3	Dream sleep (REM) occurs more in the second half of the night.	53 (75.7)
4	Sleeping longer on weekends is recommended as a regular practice to make up for loss of sleep during the work week.	50 (71.4)
5	Newborn infants spend about 16-18 hours per 24-hour period sleeping.	60 (85.7)
6	The report of insomnia is twice as common in older men than in older women.	24 (34.3)
7	A young (preadolescent) child who regularly has trouble getting to sleep at night should be allowed to sleep later in the morning.	32 (45.7)
8	The typical age of symptom onset for narcolepsy is 40 years or older.	24 (34.3)
9	The ability to sleep increases in persons above 50 years of age.	47 (67.1)
10	Slow-wave sleep is more prominent in the second half of the night.	27 (38.6)
11	The amount of slow-wave sleep increases in persons above 50 years of age.	20 (28.6)
12	Episodes of sleepwalking tend to occur in the last third of the night.	19 (27.1)
13	Episodes of REM sleep tend to lengthen throughout the night.	32 (45.7)
14	Periodic limb movements during sleep are typically decreased in REM sleep.	28 (40.0)
15	Hyperactivity in children can be exacerbated by inadequate sleep.	55 (78.6)
16	In alcoholics in recovery, sleep normalizes within 1 month of alcohol abstention.	11 (15.7)
17	Daytime napping is recommended for patients with difficulty initiating sleep.	42 (60.0)
18	Weight loss is often indicated in the treatment of primary snoring or mild obstructive sleep apnea.	58 (82.9)
19	Slow-wave sleep is enhanced following daytime exercise.	42 (60.0)
20	Chronic bedwetting in children responds to treatment with anticholinergic drugs.	6 (8.6)
21	Nightmares are more common within the first 2 hours of sleep.	19 (27.1)
22	Heart rate, respiration, and blood pressure are more variable during REM sleep compared to non-REM sleep.	38 (54.3)
23	Antihypertensive drugs (e.g., beta-blockers) may cause sleeping difficulties as a side effect.	36 (51.4)
24	Early morning awakenings in the elderly are often associated with the timing of their biological rhythms.	49 (70.0)
25	Alcohol can be beneficial in reducing the effects of jet lag.	43 (61.4)
26	Night shift workers are more likely to fall asleep on the job compared to employees with regular, daytime hours.	48 (68.6)
27	Episodes of sleepwalking commonly occur during REM sleep.	13 (18.6)
28	Menopausal women are at higher risk for developing symptoms of sleep apnea compared to premenopausal women.	43 (61.4)
29	Irregular sleep scheduling can increase the incidence of sleepwalking in children.	45 (64.3)
30	Symptoms of narcolepsy are related to seizure activity in the brain.	5 (7.1)

questionnaire. A total of 12 D3 students and 13 D4 students were in the high knowledge group; Chi-square test revealed no statistically significant correlation. Mean percentage scores for study participants based on demographic information can be referenced in Table II.

ASKME questionnaire subtopics

The ASKME questionnaire distributed for this study included 4 subtopics in sleep medicine: sleep physiology and sleep stages, sleep and age, sleep disorders and abnormalities, and sleep and medical conditions/pharmacology. The mean correct answers per subtopic can be referenced in Table III.

DISCUSSION

The results of this study demonstrate a low level of knowledge regarding sleep medicine in both D3 and D4 students. While the findings of this study align with similar previous findings in other countries, this is cause for concern. However, D4 students performed better on the ASKME questionnaire than D3 students (51% and 48.1% correct, respectively), the difference in correct answers was not statistically significant. This finding indicates that the progression through dental school education does not substantially increase the understanding of sleep medicine knowledge.

Statistically significant differences were found between genders and the mean number of correctly answered questions on the ASKME questionnaire. Few previous studies have directly compared the correlation between gender and sleep medicine knowledge, making comparisons difficult. Despite these differences between predictor variables and ASKME questionnaire scores, none of the studied demographic groups had a mean score of $\leq$60%, with most participants being placed in the low knowledge category With 67.1% of participants answering less than 18 questions correctly, the results of this study demonstrate that proactive changes should be made to the dental school curriculum

Table II. Participant demographic information and the mean percentage of each survey question answered correctly

		Study participants (N = *70*)	
		# of participants	*Mean % correct*
Age	**<26 years**	49	50.8%
	≥27 years	21	46.2%
Current dental year	**D3**	40	48.1%
	D4	30	51.0%
Gender	**Male**	33	44.1%
	Female	35	54.8%
	Prefer not to say	2	40.0%

to ensure that all dental students receive a holistic education, including on topics of sleep medicine.

A similar 2021 study by Alkhader and Sadeh[15] in Jordan resulted in a mean ASKME questionnaire score of 40%, compared to the 49.3% score in our study. Similarly, Almohaya et al.[5] reported in their 2013 study in Saudi Arabia that >95% of the participating medical students scored <60% correctly, while a 2017 study by Saleem et al.[17] that used similar metrics reported a mean score of 48% correct. A study in Croatia noted that "medical students, postgraduate physicians, and specialists showed poor knowledge but positive attitude toward sleep medicine."[18] These studies play a crucial role in evaluating knowledge in sleep medicine and demonstrate that this lack of understanding is a problem in many countries around the world. The consistency in the lack of sleep medicine knowledge among dental students at different institutions and across continents strengthens the argument for increased sleep medicine education in healthcare programs.

Dental school curricula

The curriculum for Case Western Reserve University dental school, the location of this study, does not include any time spent on sleep medicine education, making the mean hours of sleep medicine education 0 which is significantly less than the already insufficient national average. A review of the dental school curriculum found no mention of sleep medicine, OSA, sleep-disordered breathing, or oral appliance therapy. Qualitative interviews with D3 and D4 dental students at CWRU revealed that some of them did receive lectures on sleep apnea, but this is not part of the official curriculum and thus subject to change year-to-year.

Table III. The mean percentage of each question answered correctly by subtopic

Subtopic	*# of questions*	*Mean % correct*
Sleep physiology and sleep stages	9	50.2%
Sleep and age	5	59.7%
Sleep disorders and abnormalities	14	45.1%
Sleep and medical conditions/ pharmacology	2	67.1%

The findings of this study demonstrate an overall lack in knowledge regarding sleep medicine. Our analyses of the subtopics found that dental students had the least knowledge regarding sleep disorders and abnormalities. Only 5 total participants (7.1%) correctly answered "false" to the question "symptoms of narcolepsy are related to seizure activity in the brain." Students also significantly lacked knowledge on the topic of sleep physiology and sleep stages. While our study found that dental students had the most knowledge of sleep and medical conditions/pharmacology, it should be noted that this category contained the least amount of questions, making it an important topic for additional research. Overall, 25% of the answers submitted for this survey were "I don't know," which further demonstrates a lack of knowledge as well as confidence by dental students at Case Western.

Strengths of this study include the use of the ASKME questionnaire, a validated measure of sleep medicine knowledge, which allows for comparison across multiple studies and between countries. Additionally, this study had participation from both third- and fourth-year dental students, allowing for researchers to gain insight into sleep medicine knowledge among those actively engaged in patient care. This study is one of the few existing studies on sleep medicine knowledge in American dental students, adding to the field of research and providing valuable insight into the gap in education in US dental schools. Limitations of this study include small sample size, self-reported data, and our inability to assess how often participants were guessing the answers due to the setup of the ASKME questionnaire.

Possible solutions

Dental schools in America should incorporate more sleep medicine education into its dental school curriculum. These topics can be included in both preclinical and clinical years.[13] When implementing these changes to the curriculum, dental schools should focus on the

areas in which students lack the most knowledge: sleep disorders, abnormalities, sleep physiology, and sleep stages. Programs should aim to increase the mean number of hours spent on sleep medicine throughout their curricula.

Another possible solution is to offer a training course on sleep disorders for all dental students. As demonstrated by a study in China, 93.8% of students regarded sleep disorders as a "significant issue" after one 3-hour course.[19] This course also helped students learn how to identify specific sleep disorders such as snoring, restless legs syndrome, nightmares, and somniloquy.[19]

CONCLUSION

The results of this study show that dental students receive insufficient instruction in sleep medicine. In the screening, diagnosis, and treatment of sleep problems, dentists play a significant role. Dental schools should take a proactive approach to integrating more sleep medicine education into their curricula to ensure that all students have knowledge in this area.

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DECLARATIONS OF INTEREST

The authors declare no conflicts of interest, financial or otherwise.

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CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Mohamed Eltoukhi: Writing – original draft, Methodology, Formal analysis, Data curation. **Nadeen Haj Ahmad:** Project administration, Conceptualization. **Dale Baur:** Writing – review & editing, Validation.

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